Dual Excitation Multiphase Electrostatic Drive

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***Abstract*– A novel electrostatic drive technology named *Dual Excitation Multiphase Electrostatic Drive (DEMED)* was presented. A basic DEMED consisted of two plastic films in which 3-phase parallel electrodes were embedded and was driven by a 3-phase ac excitation to the electrodes. Static characteristics of DEMED were calculated and tested and the results agreed very well.**

Three prototype motors of DEMED were fabricated using commercially available technique. The first prototype consisted of a single slider and stator and generated a linear motion with a slider's motion range of about 5mm. It weighed 7g and generated a power of 1.6W and a thrust force of 4.4N. The second prototype consisted of 50 layer stack of linear motors, summing their outputs. It weighed 3.6kg and generated a propulsive force of 310N being powered with boosted commercial 3-phase electricity. The third prototype consisted of a rotor and a stator in which electrodes were arranged radially and generated rotational motion. The maximum power of 36mW was generated by the prototype weighing only 260mg for its rotor and stator.

From the results of the numerical calculation, a practical design methodology for the motor was determined. An optimal design for a motor employing currently available material and fabrication techniques is provided as an example. Analyses predict that force generation over the interfacial area between the slider and stator of this motor would be 3900N/m^2.

I. INTRODUCTION

Until now, most electromechanical systems like those found in industrial robots have employed electromagnetic motors because of ease-of-operation and large output power. However, many leading-edge high-performance electromechanical systems demand stronger, lighter-weight, and more compact actuators. For example, for robotic arms to possess capabilities equivalent to human ones, motors with power densities as high as 1kW/kg are required, while the current standard for advanced ac servo motors is about 0.2kW/kg. Because the technology for these motors has completely matured after a century of progress, the prospects of it fulfilling these increasing needs are unlikely.

Comparatively, sub-millimeter scale electrostatic motors are known to be capable of producing more power and force than electromagnetic ones with the same dimensions, and as a result, various micro-scale electrostatic motors have been developed in the past several years[1, 2, 3]. It follows that this small motor technology can be applied to not only microscopic machines but also regular sized industrial machinery by linking a large number of micro motor units in parallel and summing their output forces[2, 4, 5]. To realize high-power generation using such an electrostatic motor linkage, the authors developed a linear surface-drive variable capacitance motor, named "Dual Excitation Multiphase Electrostatic Drive (DEMED)." This paper presents principle and calculation of DEMED and introduces various types of prototype motors which employ the novel electrostatic drive technology, and discussion about optimal design of DEMED is made.

II. OPERATION METHOD

A fundamental DEMED unit consists of a pair of plastic films, slider and stator, as shown in Fig. 1. Each film is equipped with 3-phase parallel electrodes to which 3-phase alternating voltages are applied. As depicted in Fig. 2, this excitation generates running potential distributions in the two films. The potential difference causes a resultant electrostatic force between the films as shown in Fig. 3. If the phase difference between the two distributions is kept relatively small, a repulsive force is generated. The normal component of the repulsive force acts to reduce the frictional force between the slider and the stator, and the tangential one propels the slider along the stator's surface.

This dual excitation is distinguished from excitation used in normal variable capacitance motors, which typically is applied

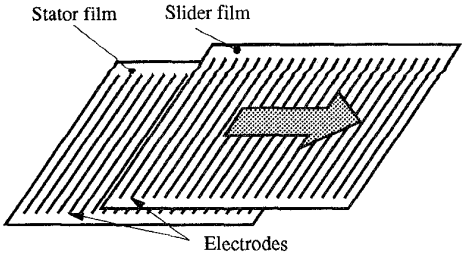


Fig. 1 Schematic view of DEMED. The electrodes are connected into three phases.

[1] This work was done mainly at *Higuchi "Ultimate Mechatoronics" Project* of Kanagawa Academy of Science and Technology.

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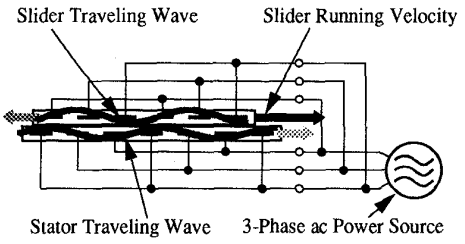


Fig. 2. Schematic cross sectional view of basic DEMED. Two potential distribution arrond electrodes are indicated by bold lines.

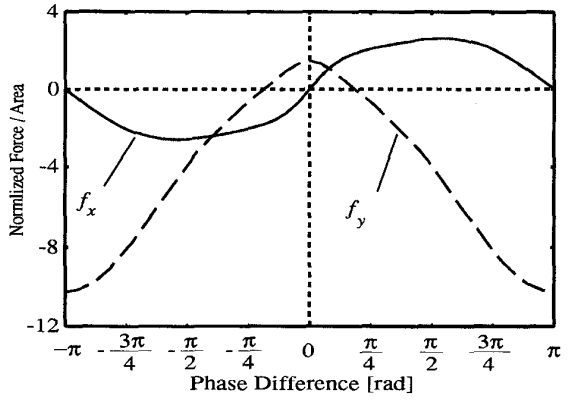


Fig. 3. Forces generated by field interaction of two completely sinusoidal potential distributions. This result is obtained in analytical way[6], not numerical.

to electrodes in the slider or stator, but both. Because of this unique excitation, DEMED can generate a repulsive force between its slider and stator reducing frictional force that hinders the sliders smooth movement.

III. NUMERICAL ANALYSIS OF STATIC CHARACTERISTICS

In this section, static characteristics of a real DEMED motor are calculated by a surface charge method, and the accuracy of the results is confirmed by experimental measurement.

For the purpose of precise positioning and static force measurements, a special test motor incorporating a pair of 5mm thick rigid plastic boards as slider and stator were built, while thin flexible films would be used in a regular device. On the slider and stator, 3-phase electrodes were fabricated at 0.5mm intervals and an insulating plastic film was adhered with epoxy resin. In order to prevent breakdown in the gap between the slider and the stator, the slider and stator were immersed in dielectric fluid which was sold commercially as *Fluorinert* by 3M.

For ease-of-calculation of the test motor's static characteristics, the following assumptions were made:

a) Each electrode is enough long to use a two dimensional field model.

b) A zero potential ground plane is positioned at a distance of two orders of magnitude greater than the pitch of the electrodes. This allows the absolute potential of each electrode to be determined with negligible effect on the calculation.

c) The number of electrodes is large, making the effect of electrodes near the edges of the motor negligible.

Fig. 4 shows the static characteristics of the calculated propulsive force between slider and stator as a function of both the phase of applied ac and position of the slider.

In order to confirm the accuracy of this estimation, static characteristics of a real motor were measured experimentally. The slider and stator were positioned with optical stages and the force was measured with a load cell. Fig. 5 shows the measured static characteristics which agree very well with the numerical computations.

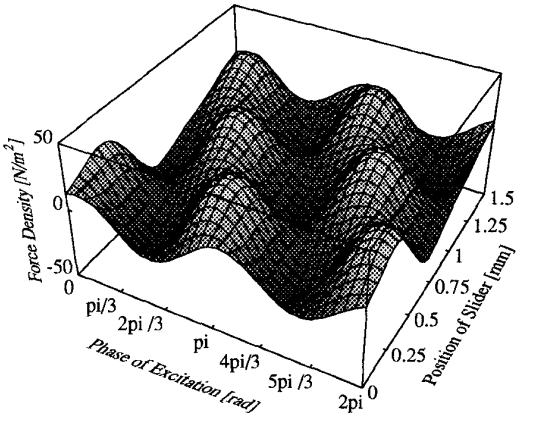


Fig. 4. Calculated static force characteristics.

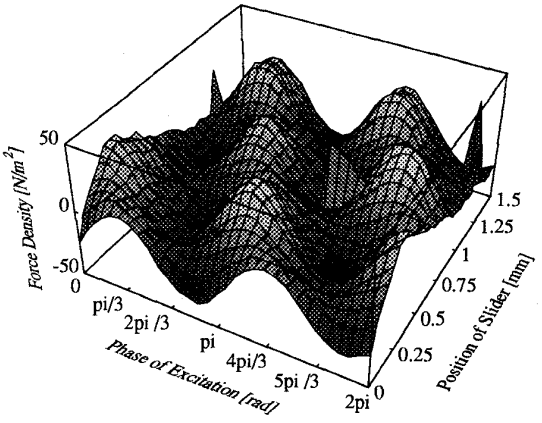


Fig. 5. Measured static force characteristics.

IV. ANALYSIS OF DYNAMIC CHARACTERISTICS

In this section, we assume that the dynamic characteristic of the motor is dominated by viscous resistance. Thus, the dynamic propulsive force of the motor can be expressed as

$$f_{DY} = f_{ST} - f_{VR} \tag{1}$$

where f_{ST} and f_{VR} are static propulsive force and viscous resistance force.

Here, the viscous force is defined by

$$f_{VR} = \frac{\rho k}{g} u \tag{2}$$

where ρ, k, g and u are density of dielectric fluid injected

between the slider and stator, kinematic viscosity, gap length between the slider and the stator, and velocity of the slider, respectively. Hence, dynamic characteristic of the propulsive force is

$$f_{DY} = f_{ST} - \frac{\rho k}{g} u \tag{3}$$

From (3), the output power of the motor can then be found to be

$$W = f_{DY} u = \frac{g f_{ST}^2}{4k\rho} - \frac{k\rho}{g}\left(u - \frac{g f_{ST}}{2k\rho}\right)^2 . \tag{4}$$

W reaches the maximum value of

$$W_{\max} = \frac{g f_{ST}^2}{4k\rho} \tag{5}$$

at a velocity of

$$u\big|_{W = W_{\max}} = \frac{g f_{ST}}{2k\rho} . \tag{6}$$

V. TEST OF BASIC PROTOTYPE

A thin and flexible motor was built to measure basic performances of DEMED.

A. Fabrication

The slider and stator shared the same structure. In each element, three blocks of electrodes (one large central block and two narrow blocks which straddled the center) were fabricated as shown in Fig. 6. Each block had 312 lines of electrodes with a pitch of 320μm. Fig. 7 is a cross sectional view of the motor. The films were processed in the same manner as flexible printed circuit board. For preservation of the gap width, a small number of 20μm diameter glass balls were sandwiched between the two films. In order to prevent breakdown, *Fluorinert* is also injected in the gap. The total weight of the slider, the stator, fluid was approximately 7g.

B. Static Characteristics

Static propulsive force was simply measured by weight lifting experiments. The maximum weight that could be lifted determined a measured propulsive force. The frequency applied to the electrode was 1Hz, yielding a very slow corresponding synchronized speed of 1.9mm/s. Fig. 8 shows the measured

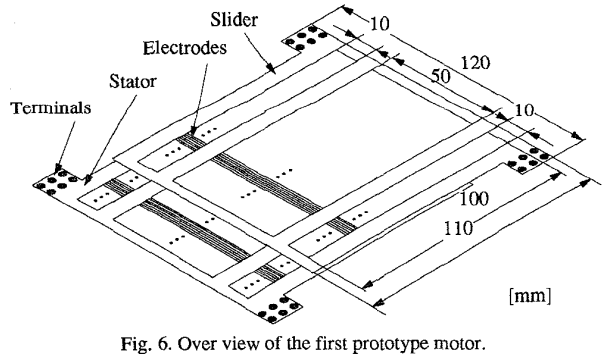


Fig. 6. Over view of the first prototype motor.

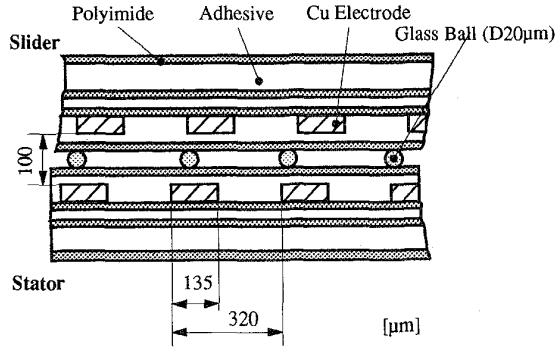


Fig. 7. Cross sectional view of the prototype motor. The cover films are adhered with epoxy resin.

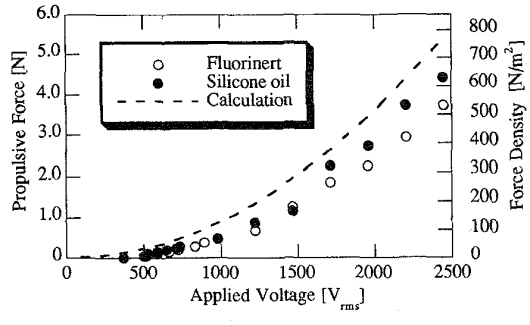


Fig. 8. Relationship between propulsive force and excitation voltage. Calculation was done assuming Fluorinert was used.

force of the motor using two dielectric fluids, Fluorinert and silicone oil, which had relative permittivities of 1.9 and 2.5 respectively. The propulsive force when silicone oil was used reached the top value of 4.4N. From this value, we calculate force per unit area as 630N/m^2. The maximum of applied voltage was limited by breakdown between two neighboring electrodes.

A result of calculation assuming Fluorinert is injected between the slider and stator is also displayed in this Fig. 8 by a broken line. Generally, electrostatic forces are proportional to applied voltage. These analytic characteristic behaviors agreed with measured thrust forces. However, there is disparity between the analytic and experimental results in terms of magnitude. For example, a maximum force of 540N/m^2 was found experimentally while 770N/m^2 was predicted analytically. This difference seems to result from incorrect estimation of area over which electrostatic forces were generated. Although the calculation assumes that the electrostatic forces are generated over the whole surface where electrodes were fabricated, the reality was that they could not be because the slider and the stator films were not perfectly flat; strong curl of films was observed.

C. Dynamic Characteristics

The same apparatus used for static force measurement was also employed in order to measure dynamic characteristics.

Since the slider's motion range was limited, pull-out characteristics could not be measured. However pull-in characteristic measurement was possible by starting the motor, which was connected to some weights, and testing whether the operation got into synchronized or not. Fig. 9 shows a characteristic of pull-in force and output power. When the operation speed is lower than 0.5m/s, the calculation agrees with the measured pull-in force well. The difference at higher speed is that between pull-in force and pull-out ones.

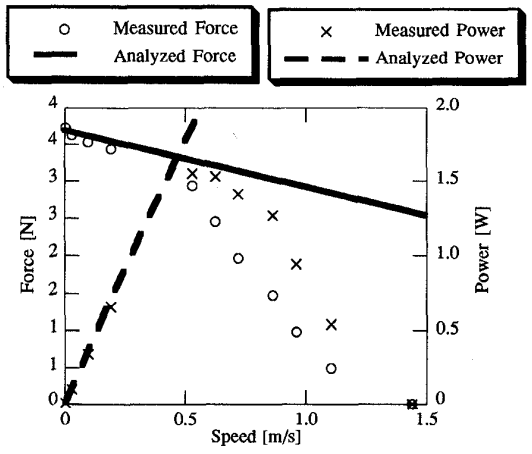


Fig. 9 Reletionships between force and speed and between power and speed. Power was calculated from force and speed.

VI. MULTIPLE LAYER MOTOR

The first prototype motor achieved high power per mass and force per mass ratios. However, its outputs themselves were not sufficient to actuate normal sized mechanical devices. In order to remove this problem, we staked many thin motors to layers.

A Fabrication

Fig. 10 shows a pair of slider and stator which were specially fabricated for a multiple layer motor. The films were also processed with flexible printed circuit board technology. Each film had parallel electrodes embedded in them at 200μm intervals. Fifty sliders and stators were stacked alternately to create a 50-layer slider-stator unit as shown in Fig. 11. Linkages between slider layers were made by eight steel pins of 3mm diameter. Similarly, linkages between stator layers were made by four pins of 4mm diameter. Between every layer, a polyimide spacer sheet with a thickness of 250μm was inserted in order to prevent the sliders and stators from squeezing each other. Finally, the alternating slider and stator layers were sandwiched by 3mm thick polyimide plastic sheets.

The entire slider-stator unit was enclosed by an acrylic plastic casing which was filled with dielectric fluid, in order to prevent dielectric breakdown. Two 4mm diameter steel connecting rods transfer the total working force of the slider-stator unit out of the casing. Fig. 12 shows a photograph of the 50-layer motor. The total weight of the motor including all the parts was 3.6kg.

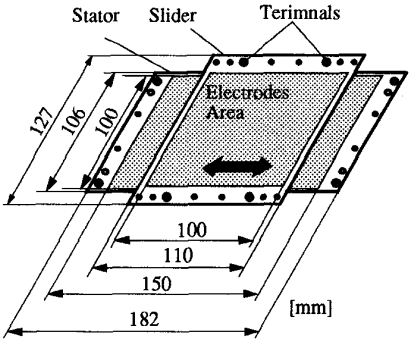


Fig. 10. Slider and stator specially fabricated for multiple layer motor.

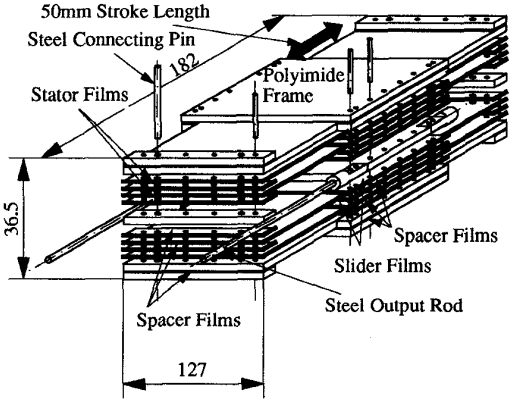


Fig. 11. Structure of 50-layer slider-stator unit.

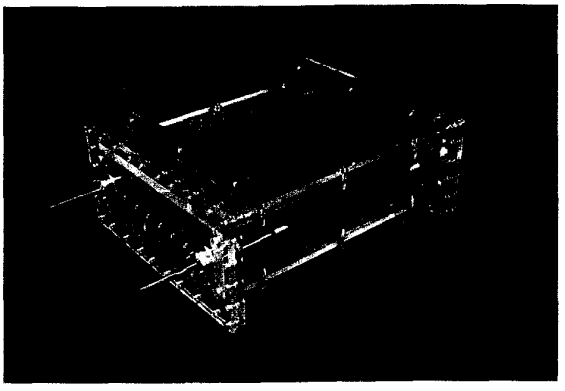


Fig. 12. 50-layere motor.

B Experimental Results

Fig. 13 shows force–voltage characteristics of the 50-layer motor and its basic single layer motor. From this result, we can find that an output of multiple layer motor was a simple multiple of that of single layer motor by the number of layers. The resultant force was also approximately proportional to the square of the voltage; and a maximum value of 310N, or 32kgf, was achieved by an excitation of $1400V_{rms}$. The maximum value was limited by dielectric breakdown between two neighboring electrodes. When the applied voltage was lower than 300V, the

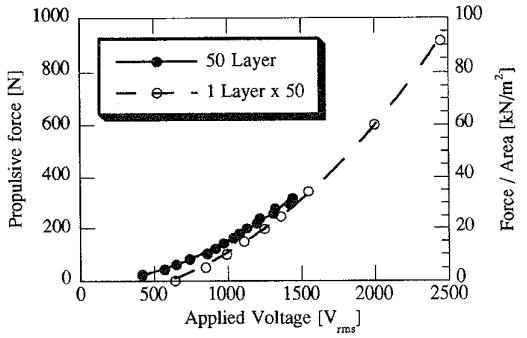


Fig. 13 Reletionship between propulsive force and applied voltage. Applied voltage was limited by breakdown between two neighboring electrodes.

motor did not overcome 3N of frictional force and fails to operate.

VII. DISK TYPE MOTOR

DEMED technology is also applicable to a motor which generates rotational motion by arranging electrodes radially.

Fabrication

Fig. 14 shows schematic views of rotor and stator for disk type motor. Each of the rotor and stator had 192 electrodes located radially at intervals of 1.875°. To the electrodes in the rotor film, electricity was supplied through the contact of the brush and slip ring as shown in Fig. 15. The rotor and stator film are processed in the same manner as the two previous prototypes and weighed 0.13g respectively.

B Experimental Results

Fig. 16 shows measured dynamic characteristics. To the electrodes a 3-phase ac of $1800V_{rms}$ was applied. When the operation speed was from 200r/min to 550r/min, analytic results and measured data agreed very well. However, when the speed is higher, they did not because of mechanical resonant of brushes and leaf springs which push the brushes onto the slip rings. Disagreement under 200r/min seems to be a result of unstable operation because of torque ripple.

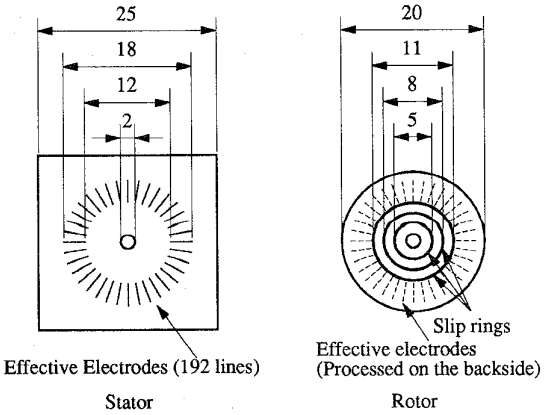


Fig. 14. Stator and rotor of the disk type motor. The rotor was processed in the same way as the stator and cut from it.

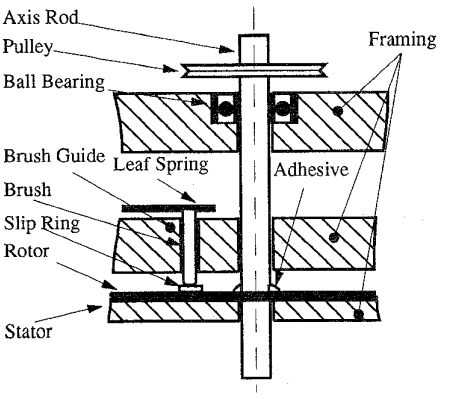


Fig. 15. Cross sectional view of disk type motor.

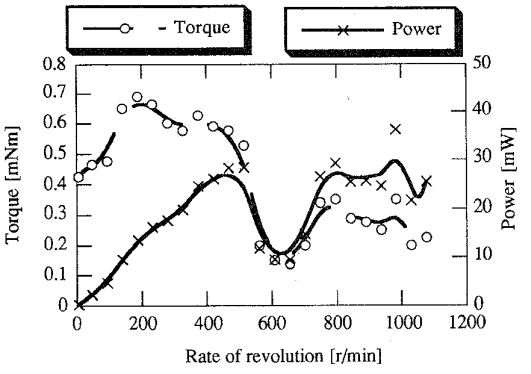


Fig. 16. Dynamic characteristic of disk type motor

The motor generated the maximum torque and power of 0.69mNm and 36mW respectively, and the power per mass of rotor and stator was calculated to be 140W/kg.

VIII. DESIGN OPTIMIZATION

In this section, maximum performance of linear type DEMED which can be fabricated using available material and technology is discussed. For the optimization, surface charge method is also employed.

A. Performance Function for Optimization

For design optimization a performance function is defined. This part describes the way to calculate the function.

Fig. 17 expresses the calculation result of static force using contour lines. If a 3-phase ac voltage is applied to the motor under a constant load, the motor's operation proceeds along the contour line which represents the same force as the load. However, if the load is so large that corresponding contour line does not exist or is closed, the load cannot be propelled. It follows that the highest open contour line represents the maximum possible propulsive force and can be used to estimate force generation performance of the motor. Since searching for this line is too computationally demanding, an approximate linear solution, or "synchronization path" as shown in Fig. 17 with a bold line, can be calculated. Each synchronization path approximates a contour, and the force it represents is given by

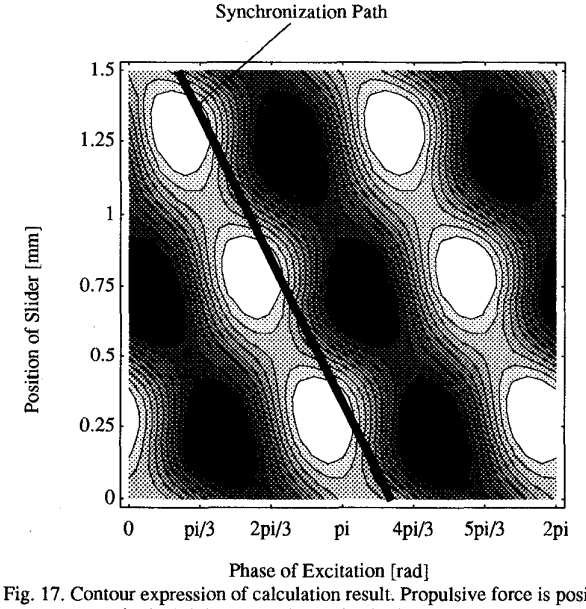


Fig. 17. Contour expression of calculation result. Propulsive force is positive in the bright area and negative in the dark area.

the minimum along its line. In order to find the maximum possible thrust force, all synchronization paths at small discrete intervals were tested.

B. Optimized Parameters

Fig. 18 is a schematic of the model used for motor optimization. For simplicity, it was assumed that the slider and stator films have uniform permittivity. Parameters subject to optimization are: electrode pitch and width, insulating cover film thickness and permittivity, slider and stator gap, and permittivity of gap fluid.

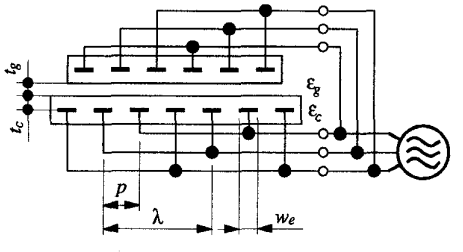


Fig. 18. Model of optimization. ε_g and ε_c are representing permittivities of gap between slider and stator and material of slider and stator respectively.

C. Pitch of Electrodes

Generally, electrostatic forces are proportional to square of electrostatic field. Therefore, the only way to obtain a large electrostatic force is to generate a strong electrostatic field. In electrostatic motors, this means the application of high voltages or a reduction of size by decreasing the electrode pitch. However, since the maximum electric field is limited by breakdown strength of insulating materials employed in the motor, two differently pitched motors that share a similar design and consist of the same materials generate equivalent propulsive forces when the respective maximum non-breakdown voltages are applied.

Although electrode pitch does not effect propulsive force of motor, the necessary supply voltage is reduced. Therefore, smaller pitched electrodes are preferable.

D. Width of Electrodes

Electric field strength between two neighboring electrodes, as shown in Fig. 19 can be approximated by

$$E = \frac{\Delta V}{p - w} \tag{7}$$

Since E cannot exceed breakdown field of dielectric material insulating the two electrodes, E_{brk}, the maximum potential difference that can be employed in practice is described by

$$\Delta V_{\max} = (p - w) E_{brk} \tag{8}$$

It follows that different voltages must be applied when comparing two designs of motors with different electrode widths. A normalized amplitude V_{amp} can be expressed as

$$V_{amp} = \frac{\Delta V_{\max}}{\sqrt{3}} = \frac{(p - w) E_{brk}}{\sqrt{3}} \tag{9}$$

Fig. 20 shows relationship between force generation performance and electrode width assuming that electrode pitch is constant. This result says that an electrode width should be around 20–30% of electrode pitch.

E. Permittivities and Thickness of Coverage and Fluid Gap

If permittivities of insulating coverage over the electrodes and fluid filled in the gap between the slider and the stator are equal, force generation of the motor is proportional to the permittivity and it is apparent that a larger value is preferable. Practically speaking, however, there are limits in the

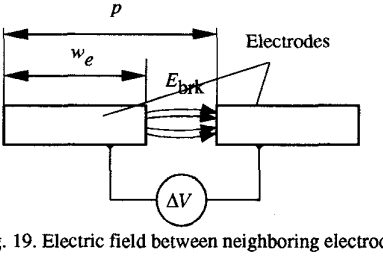


Fig. 19. Electric field between neighboring electrodes.

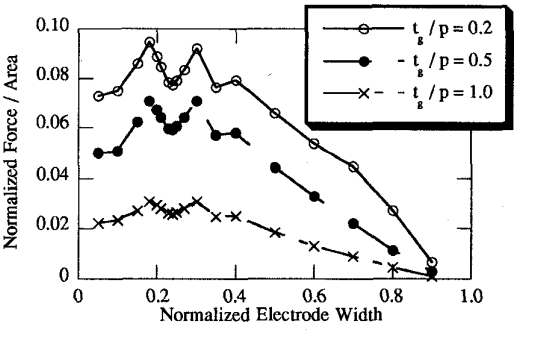


Fig. 20. Relationship between force per unit area and width of electrode. Force per area f_s is normalized by $f_s/(\varepsilon_c E_{brk}^2)$ using $\varepsilon_c = 1$. Width is normalized using w_e/p.

permittivities for materials which can be used for the motor. In particular, fluids with high permittivities are rare. Although fluids consisting of polar compounds such as water, ethanol or nitrobenzene have large permittivities, they cannot be used because their breakdown strengths are low. In this section, discussion proceeds with the assumption that the fluid permittivity is lower than that of the insulating cover.

Fig. 21 shows the relationship between force generation performance and fluid gap permittivity assuming that of coverage is constant. This result reveals that a fluid permittivity as close as possible to that of insulating cover is desirable.

Fig 22 shows the relationship between force generation performance and cover permittivity assuming that of the fluid is constant. This result also indicates that large insulating coverage permittivities are preferable.

Fig. 23 shows the relationship between force generation performance and thickness of insulating coverage. This result shows that cover film thickness should be near 10% of electrode pitch.

Fig. 24 shows the relationship between force generation performance and length of fluid gap. This result clarifies that the gap should be minimized.

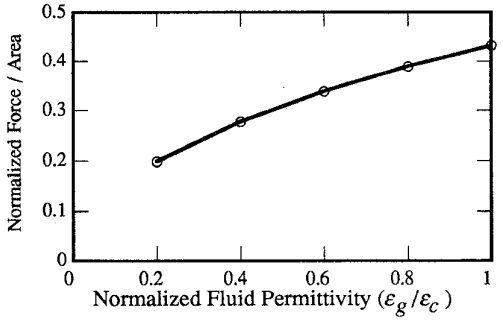


Fig. 21. Relationship betwenn force per unit area and permittivity of fluid gap. Force per unit area f_s is normalized by $f_s/(\varepsilon_c V_{amp}^2/p^2)$ using $\varepsilon_c = 1$.

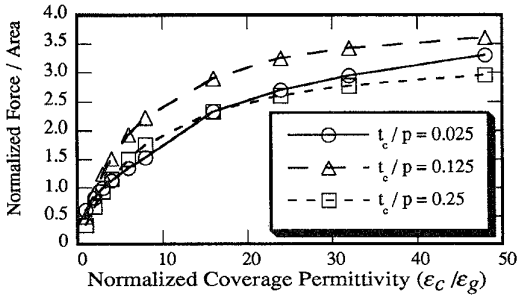


Fig. 22. Relationship betwenn force per unit area and permittivity of cover film.Force per unit area is normalized by $f_s/(\varepsilon_g V_{amp}^2/p^2)$ using $\varepsilon_g = 1$.

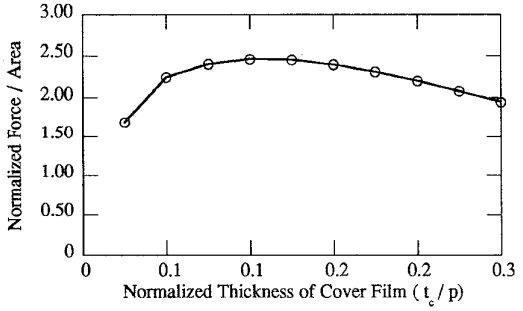


Fig. 23 Relationship betwenn force per unit area and thickness of coverage film. Permittivity ratio between fluid and coverage film is assumed $\varepsilon_c/\varepsilon_g = 10$. Force per unit area is normalized by $f_s/(\varepsilon_g V_{amp}^2/p^2)$ using $\varepsilon_g = 1$.

F. Performances of Optimally Designed Motor

Tab. I shows optimal design employing currently available materials and fabrication techniques. The electrode pitch of 200μm is equivalent to the minimum value which can be typically fabricated using a flexible print board technique. For the insulating coverage and the dielectric fluid, the motor employs PVDF and silicone oil which have relative permittivity of 13 and 2.2 respectively. The amplitude of the ac supply voltage is determined from the breakdown strength of epoxy resin normally used to adhere coverage film. According to the calculation for this optimally designed motor, a force per unit area ratio of 3900N/m^2 should be possible. From this value, maximum force per mass ratio can be roughly calculated as 40kN/kg using densities of PVDF film and silicone oil. This value means the motor can lift up a weight four thousand times as heavy as the motor itself.

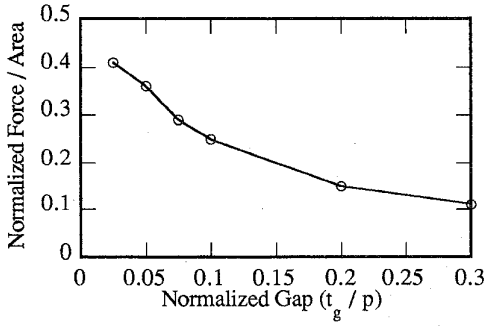


Fig. 12 Relationship between force per unit area and fluid Gap. Permittivity ratio between cover film and fluid gap is assumed $\varepsilon_c/\varepsilon_g = 10$. Force per unit area is normalized by $f_s/(\varepsilon_c V_{amp}^2/p^2)$ using $\varepsilon_c = 1$.

TABLE I
EXAMPLE OF THE OPTIMALLY DESIGNED MOTOR

Electrode Pitch	200μm
Electrode Width	40μm
Cover Film Thickness	12.5μm
Fluid Gap	10μm
Cover Film Material	PVDF
Fluid	Silicone Oil
Applied Voltage	2000V

IX. CONCLUSIONS

Dual excitation multiphase electrostatic drive, DEMED, was developed. Static characteristics of DEMED were numerically calculated and tested. For the calculation, surface charge method was used, and for the test a data measurement apparatus equipped with a specially fabricated motor which consisted of a rigid slider and stator. The results of the calculation and the measurement agreed very well. Three prototype motors were fabricated by commercially available technology and their performances reached the highest level of electromagnetic

motors. Additionally, discussion for optimal design of DEMED was made, and it showed that the motor which was designed and fabricated using currently available materials could generate a force per mass ratio of 40kN/kg. All the results in this paper proved that prospects of electrostatic motors as power sources of regular sized mechanical devices are quite likely.

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REFFERENCES

[1] L. S. Fan, Y. C. Tai and R. S. Muller, " IC-processed electrostatic micro-motor," *Proc. IEEE International Electron Devices Meeting*, Sanfrancisco, CA, 1988, pp.666-669.

[2] S. C. Jacobsen, R. H. Price, J. E. Wood, "The wobble motor: an Electrostatic, planetary-armature, microactuator," *Proc. 1989 IEEE Micro Electro Mechanical Systems Workshop*, Salt Lake City, UT, pp. 17-24.

[3] M. Mehregany, P. Nagarkar, S. D. Senturia, J. H. Lang, "Operation of microfabricated harmonic and ordinary side-drive motors," Proc. 1990 IEEE Micro Electro Mechanical Systems Workshop, Napa Valley, 1990, pp. 1-8.

[4] S. Egawa and T. Higuchi, "Multi-layered electrostatic film actuator, *Proc. 1990 IEEE Micro Electro Mechanical Systems Workshop*, Napa Valley, CA, 1990, pp. 166-171.

[5] T.Niino, S.Egawa, N.Nishiguchi and T.Higuchi, "Development of an electrostatic actuator exceeding 10N propulsive force," *Proc. 1992 IEEE Workshop on Micro Electro Mechanical Systems*, Travemünde, Germany, 1992, pp. 122-127.

[6] T. Niino, S. Egawa and T. Higuchi, "Ac drive electrostatic linear motor," *Proc. IEEJ 1995 International Power Electronics Conference*, Yokohama, Japan, 1995, pp. 1696-1701.